

CLICK-A-THON INDIA 2026 • SONYLIV • CLICKHOUSE + CLICKSTACK

Count the people who are actually watching.

Not sessions that are open. Not apps that are running. Viewers whose screen is lit — from start/end events and one-minute heartbeats.

7,000,000

EVENTS, UNSEEN DAY

23,324

PEAK CONCURRENT

3,201,716

MINUTES RECONCILED

0

MISMATCHED

WHAT WE MEASURED — AND WHAT IT FORCED

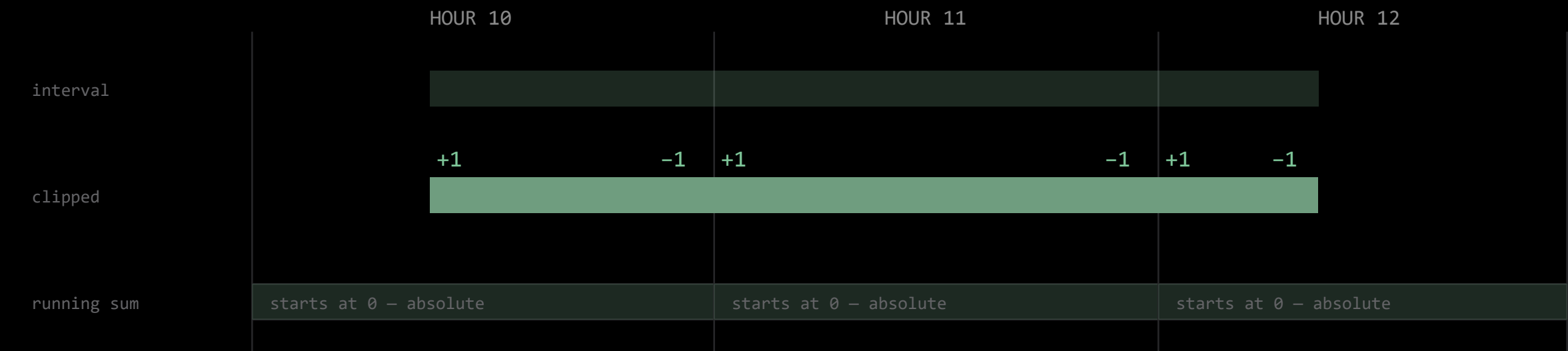
Heartbeats **survive** a pause. That one fact drove the design.



So gap detection — the obvious mechanism — **cannot see a paused viewer at all**, and a model built on it counts them as watching while looking perfectly healthy. We exclude pause explicitly, on a second signal. **This is the measurement, not an assumption.**

THE MODEL

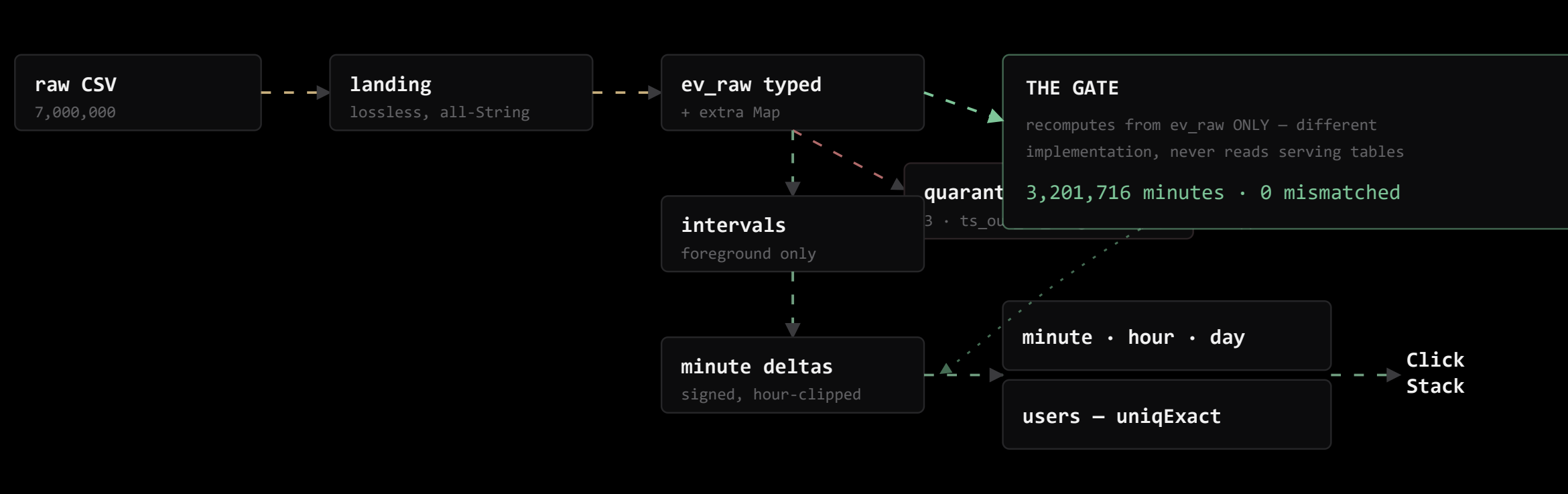
Two numbers per interval, clipped to every hour it touches.



139,925 rows describe 3.2 million minutes. A query for one hour reads one hour of deltas — never the history before it.

ARCHITECTURE

Every number traces to a raw event.



RUNTIME PREPROCESSING — EVERY ROW GETS A TERMINAL STATE

Nothing is dropped silently.

landed as text		7,000,000 — lossless
cast rejects	0 — every column typed cleanly	
quarantined	3 — <code>ts_out_of_range</code> , kept in a table, not discarded	
accepted to the model		6,999,997
exact replays collapsed	24,964 → 6,975,033 derivation rows	

LAND FIRST, TYPE SECOND All-String, header-aware. A malformed row costs a row, not the file — and a column we have never seen survives instead of being dropped at the door.	QUARANTINE, NOT DELETION Unusable rows go to a table with a reason. Three timestamps out of range. They are countable and inspectable — a judge can ask what we excluded and get an answer.	TYPE-AWARE, NOT BLANKET CLEANUP 12 case-drifted Pause controls canonicalised. Identifiers, catalogue prose and raw filter values are not rewritten — normalisation happens at query time, so storage stays raw.
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HOW WE KNOW IT IS RIGHT

The gate will not agree with itself.



mismatched minutes, out of **3,201,716** compared. Truth is recomputed **from raw events**, by a different implementation of the same spec — window functions against array splitting — so an error in one shows up instead of cancelling out. It never reads the serving tables.

FAILS ON SILENCE TOO

Zero rows compared is a **failure**, not a pass. A gate that cannot see its data must not succeed quietly.

11 TEST SUITES

Golden cohorts solved **outside** the pipeline, property tests against an independent interpreter, 32 sabotage-checked edge fixtures.

DETERMINISM, PROVEN

26 of 27 queries byte-identical across two ClickHouse versions, two engine families, 3 vs 10 threads.

THE RESULT • UNSEEN DAY 2026-07-31 UTC

23,324 at 11:17.

23,324 PEAK CONCURRENT SESSIONS	22,279 PEAK DISTINCT USERS • 11:16	944.70 TIME-WEIGHTED AVERAGE	11,277 BUSIEST HOUR AVERAGE
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The user peak lands **one minute before** the session peak — a real property of the data that a summed or averaged tier would hide.

PEAK IS NOT SUMMABLE Summing 19 platform peaks → 24,025 (+3.0%). Maxing them → 7,163 (−69.3%). Both wrong, opposite directions. Peaks land at different minutes.	AVERAGE IS TIME-WEIGHTED 81,621,960 concurrency-seconds ÷ 86,400. Not a mean of per-minute values.
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► LIVE DEMO – SWITCH TO THE DASHBOARD NOW

The curve, and filters that actually move it.

1. **Set the range** to `2026-07-31 00:00 → 23:59 UTC` – the unseen day. Outside it the panels are empty.
2. **Show the ACCURATE curve** – the ramp, and the peak at 11:17.
3. **Show NAIVE beside it** – same data, session-span reading. The gap is backgrounded and paused time.
4. **Apply a filter** – platform, or `video_resolution`. Watch the curve change shape, not just scale.
5. **Say the resolution point:** that column **arrived with the unseen file**. No migration. Filterable the same day.
6. **Show pipeline health** – our own ingestion and query latency, in ClickStack.

⚠ 11 of 12 filters move the curve. **Country does not** – it is the constant 'india' in all 7,000,000 rows. Say it before anyone asks.

⚠ **Latency on screen will not be 28 ms once you filter.** The unfiltered curve comes off the promoted delta tier at 28 ms. A filtered drilldown answers on arbitrary dimensions and takes ~4.4 s. Both are real and both are ours – say which path you are on rather than letting the number surprise you: **"that is the generic path, it answers on a column we never pre-aggregated."**

NEW DIMENSIONS, SAME DAY

An unknown column becomes a filter with no migration.



Unknown columns land in a **Map** and stay queryable. We show the fallback's cost rather than hide it — **that is what "handles new dimensions" means today.**

SCALE — MEASURED AT 1× / 10× / 100×

At 100× the build strains. Serving does not move.



89,850,838 events. 1.09 million sessions. Query latency is unchanged because the delta phases are **window-bound by design** — hour-clipping means a query reads one hour, whatever the history behind it.

WHAT BREAKS FIRST — NAMED, NOT GUESSED	OPTIMISATIONS IN PLACE	THE HONEST LIMIT
Interval derivation memory . At 100× it failed at default settings; spill alone did not rescue it; spill + 2 threads did. Dictionary memory stayed flat at 17 MiB — it keys on the catalogue, not the audience.	Hour-clipped deltas (flat reads) · promoted cubes per dimension subset · uniqExact state merge instead of re-scan · chunked backfill under the partition cap · dictionary join on the catalogue.	28 parts, 930 merges, slowest merge 36 s at 100×. Nothing wedged, but the build is where the next engineering goes — not the query path.

OPERATIONS — WHAT ACTUALLY BROKE

The unseen file did not go through cleanly.

102 DATES > 100 PARTITIONS

Cloud pins the limit **read-only**. Raising it returns **Code 452** unconditionally — a second failure, not a fix. Solved by chunking: **64 + 38**.

THE HALF-BUILT MODEL

A build died mid-run. Re-running without truncate **doubles** an AggregatingMergeTree — and looks entirely plausible. A guard now refuses to serve a partial model. **It fired for real.**

A TEST THAT STOPPED TESTING

One template silently matched nothing, turning an incremental test into a full rebuild that **converged by construction**. Caught because its sabotage half went quiet.

Every one of these was caught by something **refusing to pass**, not by someone noticing.

WHAT WE WOULD TELL YOU BEFORE YOU FOUND IT

The limits, stated.

COUNTRY IS A DEAD FILTER Constant 'india'. 11 of 12 filters move the curve.	BACKGROUNDED-STATE We use gaps, not the explicit marker. Bounded at 0.72% of hours, 0.91% at peak — measured both ways.	SESSION IDENTITY 159 ids carry two start epochs; 303 carry two users. Grouping on the session alone can merge incarnations.	THE PEAK MINUTE 11:17 leads 11:16 by six. The value is robust; the minute deserves care.
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Judges now spot-check against raw events. **Anything hidden is worth less than anything stated.**

CLOSE

Foreground-only means foreground-only.

A fast wrong answer scores zero. So the model is small, the gate is independent, and every number names the policy that produced it.

23,324 PEAK • 2026-07-31 11:17	0 MISMATCHED, OF 3.2M MINUTES	11/12 FILTERS MOVE THE CURVE	10-210ms SERVED ON CLOUD
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Full detail — lineage, algorithm, model, query path, evidence:

<docs/artifacts/2026-08-02-solution-atlas.html>